Integrated Personnel and Pay System – Army (IPPS-A)

Data Conversion and Interfaces Plan (DCIP)

W15QKN-15-C-0049-DCIP-R3-1.4-20191118

Version 1.4

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Contract No.: W15QKN-15-C-0049
IPPS-A Data Item No: 08-001

18 November 2019

REVISION HISTORY

Version	Submitted Date	Description	Author
1.0	10 MAY 19	Initial submission for Release 3	C. Smith
1.1	10 JUN 19	Revised Submitted date for Version 1.0 to match actual date. Addressed PMO comments in CS_CDRL_08-001_DCIP_R3_v1.0_20190510	R. Watson C. Smith
1.2	24 SEP 19	Addressed PMO comments in CS_CDRL_08-001_DCIP_R3_v1.1_20190610	J. Underly
1.3	22 OCT 19	Updated for CDR.	J. Underly
1.3	18 NOV 19	Updated for CDR comments in CS_CDRL_08-001_DCIP_R3_v1.3_20191022.	J. Underly

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1 INTRODUCTION AND PURPOSE

1.1 Introduction

The Army currently relies on more than 60 separate legacy information systems to provide personnel and pay management for all components (Active, Guard, and Reserve). These systems were designed at different times, reside on different hardware platforms, use a variety of in-house and commercial software, and vary widely in their functionality.

To consolidate these systems, the Army is developing and implementing the Integrated Personnel and Pay System – Army (IPPS-A), a Web-based tool, available 24 hours a day, accessible to soldiers, Human Resources (HR) professionals, Combatant Commanders, personnel and pay managers, and other authorized users throughout the Army.

Team CACI has been selected as the System Integrator (SI) to ensure successful development and delivery of the envisioned IPPS-A.

1.2 Document Purpose

The purpose of this Data Conversion and Interfaces Plan (DCIP) is to outline the pertinent considerations for the implementation of PeopleSoft Human Capital Management (HCM) for IPPS-A Increment II releases. This conversion and interfaces approach describes strategies and processes used to plan, manage, and execute conversion and interfaces of the Army legacy data into and from IPPS-A PeopleSoft HCM.

This data conversion and interfaces strategy document proposes the following:

- Define data conversion and interface scope.
- Provide a guideline for the data conversion and interface processes, used in conjunction with data mapping activities to correlate existing Army legacy data with the PeopleSoft HCM data structures.
- Outline a high-level approach for conversion of required data from the current Army legacy environments to IPPS-A and a high-level process for persistent interfaces from/to IPPS-A.
- Lay out the data conversion, quality, and control process needed to load, reconcile, and confirm the conversions.
- Detail how IPPS-A integrates with external Army systems, after data conversion is completed, and outline the high-level approach for maintaining persistent interfaces with these systems.

1.3 Document Overview

The Data Conversion and Interface Plan (DCIP) describes the strategy and approach for IPPS-A to convert data and integrate with external systems.

The IPPS-A Team reviews the DCIP as part of each release, when there are changes in project scope, when the program identifies new risk events, or when implementing process improvement recommendations.

The SI Data Integration Lead is the approval authority for this document and coordinates document changes with the SI IPPS-A Program Manager to ensure compliance with the IPPS-A program practices and policies.

1.4 Reference Documents

Table 1-1 lists key reference documents for executing the IPPS-A program.

Table 1-1 IPPS-A Reference Documentation

Document Title	Date
CACI Operational Excellence and CACI Delivery+ Framework	Varies
Continuous Process Improvement Plan V1.0	05 AUG 11
IPPS-A Contract W15QKN-15-C-0049, Mod 19	26 APR 17
IPPS-A Technical Proposal	03 MAR 16
Proposals and Engineering Change Proposals (ECP)	Varies

2 ORGANIZATION AND MANAGEMENT

The Project Management Plan (PMP) provides the organization chart for the IPPS-A program. The Data Team, supported by the Infrastructure, Application and Integration and Testing teams, is responsible for data conversion and interface development for IPPS-A.

2.1 Roles and Responsibilities

Table 2-1 defines the roles and responsibilities for the data conversion and interfaces effort.

Table 2-1 Data Conversion Roles and Responsibilities

Role	Responsibilities
System Architect	The Systems Architect is responsible for ensuring all organizational software engineering policies and processes institute and execute throughout the IPPS-A program life. This includes: ▪ Ensuring integration of appropriate software engineering procedures, methods, and tools into the program's defined software process ▪ Developing and maintaining the IPPS-A software development life cycle methodology and processes ▪ Managing the integration/alignment of data conversion efforts ▪ Ensuring all members of the engineering group are trained to perform their software development activities
Data Lead	The Data Lead is responsible for the overall data solution for IPPS-A implementation. This includes: ▪ Providing guidance, direction, and task assignments to other developers ▪ Maintaining consistency across data and interface work products, including plans, process descriptions, allocated requirements, requirements, design, code, test plans, and test procedures ▪ Ensuring that data measurements and status are developed, analyzed, and reported to determine the functionality and quality of the work products ▪ Reviewing the development activities on both a periodic and event-driven basis
Conversion and Interface Leads	The Conversion and Interface Leads are responsible for the overall design and development of all data conversions and interfaces for IPPS-A implementation. This includes: ▪ Providing guidance, direction, and task assignments to other developers ▪ Ensuring data on defects identified in peer reviews, document reviews, and unit testing are collected and analyzed according to the program's defined verification process ▪ Ensuring measurements and status are developed, analyzed, and reported to determine the functionality and quality of software work products ▪ Reviewing software development activities on both a periodic and event-driven basis ▪ Ensuring test environments are accurate and maintained ▪ Preparing documents and/or software code required to establish the design baseline, developmental configurations, test baselines, and ultimately the product baseline(s)

Role	Responsibilities
Technical Infrastructure Team	The Technical Infrastructure Team is responsible for ensuring that IPPS-A infrastructure is operational for all pre-production, production, and COOP environments. Responsibilities include: ▪ Maintaining Development, Testing, and Production environments ▪ Administering databases ▪ Deploying releases ▪ Establishing, maintaining, and troubleshooting interface and user connectivity ▪ Monitoring, troubleshooting, and improving system performance ▪ Managing system availability ▪ Performing capacity planning ▪ Managing continuity of operations planning and execution
Data Team	The Data Team is responsible for the design and development of all data conversions and interfaces for IPPS-A implementation. Responsibilities include: ▪ Creating/updating requirements, solution designs, and configuration documentation ▪ Creating/maintaining detailed design for reports, interfaces, conversion, extensions, forms, and workflow ▪ Developing and maintaining software ▪ Conducting peer reviews ▪ Designing and conducting unit testing ▪ Developing and executing test cases
Testing Team	The Testing Team is responsible for verifying that the IPPS-A solution meets the defined requirements. Responsibilities include: ▪ Defining/executing test processes via the Test Management Plan ▪ Defining requirements for test environments and coordinating maintenance of conversion test data ▪ Planning Integration, System, and Acceptance Testing; developing release-specific test plans for data conversions ▪ Overseeing development and maintenance of test cases ▪ Executing tests and reporting on test results for data conversions

3 DATA CONVERSION STRATEGY

The data conversion strategy provides a blueprint for the activities needed to accomplish the conversion of data from external systems to IPPS-A PeopleSoft HCM. The data conversion methods developed for Release 2 (R2) will be leveraged and expanded to support Release 3 (R3).

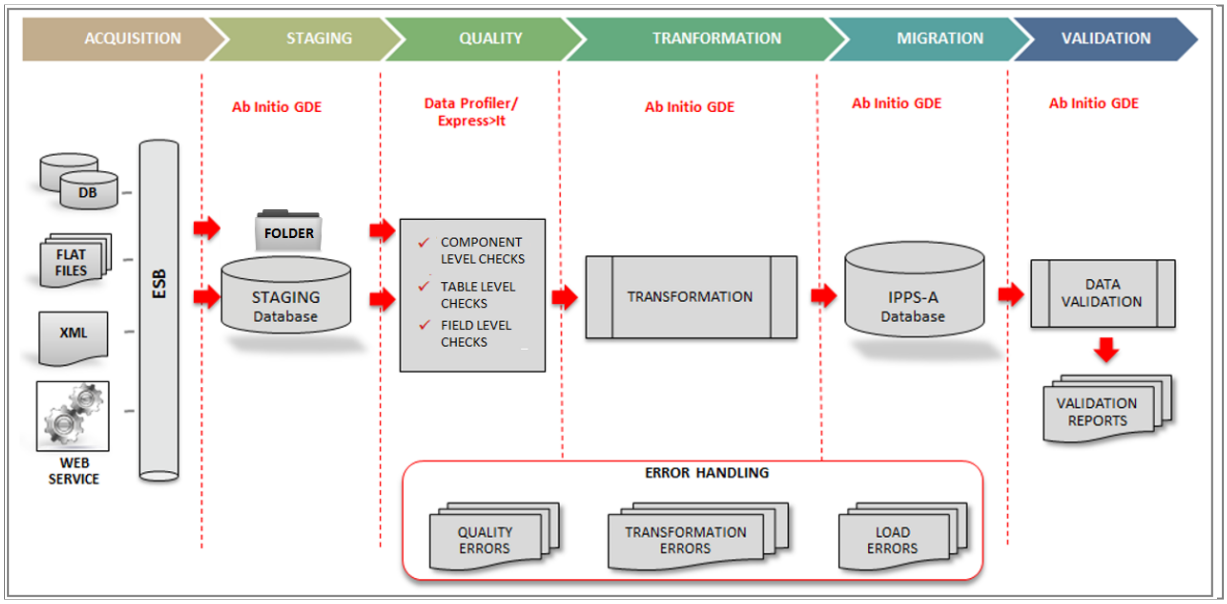
3.1 Data Conversion Scope

The general scope for data conversion is defined by the IPPS-A's SV-8 for R3 and refined in the post CDR Basis of Estimate (BOE) Scope list. The SV-8 is a living document and changes periodically. Based on the SV-8 list of systems and CDRL 13-006 Requirements Traceability Matrix (RTM), a detailed Work Item/Feature scope is baselined for each release specifying the type of data that is consumed by IPPS-A into the underlying PeopleSoft components. The Engineering Change Board (ECB) process governs changes to data Work Items/Features.

3.2 Data Conversion Approach

The data conversion process outlined in subsequent sections is specific to Army systems to be subsumed as identified on the IPPS-A's SV-8 in R3. See Appendix D: R3 Conversion Strategy for details on the approach for transitioning R2 data to R3, sequencing of the loads, testing approach, and other system level considerations. The strategy will evolve as the program progresses through build and mock conversion iterations. The data conversion process uses Ab Initio as the primary tool to convert Army source data for IPPS-A. Figure 3-1 is a high-level diagram depicting the Ab Initio conversion process and its various components.

Figure 3-1 Data Conversion Components



Ab Initio consists of multiple modules that provide a full suite of Extract, Transform, and Load (ETL) tools. The developers will primarily work within the Ab Initio frameworks and the Graphical Development Environment (GDE) where they will generate the transformation and

loading logic, which results in Ab Initio graphs. Express>It is an Ab Initio tool that is used to profile data and to produce data quality reports. The data quality reports can be viewed within the Ab Initio Metadata Hub. These are the primary Ab Initio modules that are employed by the data team but other Ab Initio modules may be utilized as appropriate.

The data conversion process comprises six sub-processes: Acquisition, Staging, Quality, Transformation, Migration, and Validation. In addition, four pre-conversion artifacts need to be in place in some capacity to enable the conversion development process.

- Complete source data from the legacy systems in the correct data layout.
- CDRL 13-005, Interface Requirements Specification (IRS), which describes characteristics of the source system including table layouts, field types, definitions, domain values, and domain code sets.
- CDRL 07-001, Interface Design Document (IDD), which is the official mapping artifact for data conversion/interfaces and contains information on historical or one time loads, source-to-target mappings, transformation logic, domain crosswalks, default values, data quality checks, and error handling.
- The Interconnection Security Agreement (ISA), a formal agreement between the source system owners and IPPS-A. It identifies the requirements and constraints required for a conversion. ISAs are maintained by the PMO and are input for the SI to build the IPPS-A system.

IRSs and IDDs are loaded into Metadata Hub and Cameo, the repository of IPPS-A's metadata and architecture, respectively. Subsequent sections describe each conversion sub-process.

3.2.1 Acquisition

The acquisition process describes how legacy source data is received by IPPS-A. There are several methods which can be used for the handling of incoming data, both file-based transfers and transactional-based data. Expected file formats are text files (csv, fixed length records, delimited records), Oracle dump files, SQL-server export files, and XML formatted files.

3.2.1.1 Development and Test Data Acquisition

During the Development phase, data acquisition will utilize one of the following methods for the transmission of data to the IPPS-A team.

3.2.1.1.1 DoD SAFE (Secure Access File Exchange)

If available, use of the DoD SAFE is the preferred method of transferring data files up to 8GB in size. DoD SAFE will not be used for recurring transfers during testing.

3.2.1.1.2 IPPS-A ESB

Once established for R3, the IPPS-A ESB may be used for the transfer of files from source systems to DISA. This method will be used for the transfer of files larger than 8GB. Coordination with the Infrastructure team is required prior to transferring large files to DISA.

For testing of R3 conversions and integrations, the IPPS-A ESB will be the primary means of moving files into and out of the IPPS-A test environments. The details of the ESB capabilities and inbound patterns are captured in the CDRL 07-004, ESB Integration Plan.

3.2.1.2 Test and Production Acquisition

During the Test and Production phases, all inbound data is received via the IPPS-A Enterprise Service Bus (ESB) utilizing one of the inbound patterns. The ISA with each system defines the frequency, method, and format of the data received. Data acquisitions for conversions and persistent interfaces may differ in data format, naming conventions and population. For example, conversion will always require the full population (complete dataset), whereas persistent interfaces can receive full population (for IPPS-A to perform delta detection) or only delta transactions. The IPPS-A Runbook contains details on interface timing.

Data received via the ESB could span multiple formats such as Extensibile Markup Language (XML), flat files, Oracle dump files, and Web Services. Regardless of the format, Ab Initio is used to read the source data in its raw format (Oracle dumps require the use of an Oracle utility). Ab Initio can load the source data into the tables in the staging database or can process the data directly from the source format.

3.2.2 Staging

The IPPS-A Landing Zone (IALZ) capability is at the heart of the staging subprocess. In addition to staging the source data received via the ESB, all subsequent conversion sub-processes read data from the IALZ.

The IALZ includes a staging schema which is a grouping of distinct schemas and houses the source data received—one database schema per source system (as applicable). The source data staging process reads data received via the ESB and loads it into tables in the appropriate staging schema as required. Some sources can also be uploaded into an encrypted folder in its raw format.

When data sources are initially established, the format and layout is documented in Data Manipulation Language (DML) that is read by Ab Initio graphs along with the source data to ensure conformance with the IRS. If no errors are encountered during the read process, the graph then loads the data into staging tables (as applicable) for that specific source system. If errors are encountered during the reading/loading process, the cause of the error is determined. Errors at this point are organized as either data or program related. For data-related errors, the SI collaborates with the PMO Data Services Team for adjudication with the Functional Management Division (FMD) and the system owner to resolve and acquire a clean data set. DML errors are resolved by debugging, updating, and testing the DML code.

Once conformance is confirmed, the data sources are registered with Query>It. Sources do not have be moved or have data modified to be registered into Query>It. Registration is done mainly by describing source record format and any keys and indices. The resulting data source is virtualized as a table in the Query>It database.

The data sources are all federated in a single integrated relational view for subsequent analysis. With a single query data analysts can retrieve data from any combination of sources, join data from any source with any other, or migrate and store data into new sources —database tables, files, or index compressed flat files (ICFFs) as required for analysis.

The conversion schema in the staging database is used to hold intermediate staging tables and crosswalk tables, which are also registered as sources within Query>It.

3.2.3 Domains, Domain Code Sets, and Code Translations

IPPS-A will maintain a catalog of reference data domains that are relevant to its core operation and for translation to external interface record sets, as required. Domains are groups of possible values for a particular property or attribute of interest--for example, "Aviation Series". The possible values available under a domain are referred to as "master values" of that domain and they represent real-world meanings—for example, the "Aviation Series" domain might have a master value of "C-12". Also maintained under a domain are one or more domain code sets, which are collections of valid code values that are associated with the domain's master values. Domain code sets can be established for a number of reasons, such as code sets proposed by different standards defining organizations (e.g., ANSI, ISO), different code value lengths (e.g., ISO 3166-1 alpha-2 vs. ISO 3166-1 alpha-3), and independently devised code sets (e.g., "System X's Country Lookup Table").

Army G1's Table Edit Support System (TESS) is the primary source of reference data information—including domains, master values, and domain code sets and values—for the IPPS-A program. TESS is a catalog of domains and domain code sets used throughout existing Army HR systems. IPPS-A maintains an inbound interface from TESS to receive all available domain code sets and incorporates relevant reference data information from TESS interface files based on requirements identified in other interface and conversion design specifications. However, not all data trading partners are guaranteed to use TESS code sets, in which case, the trading partner must supply the list of possible values and their meaning (i.e., master values).

As part of reference data code set onboarding, regardless of whether from TESS or another source, Army SMEs must determine whether a code set encodes a domain already incorporated within IPPS-A or a new domain will be created. Furthermore, for each code value within the code set, Army SMEs must determine if the value aligns with an existing master value or a new master value should be created. By aligning code sets to domains and code set values to master values, a "cross-walk" or translation of encoded values to one another can be performed wherever required. The system integrator will support Army SMEs in supplying a catalog of domains, master values, code sets, code values already incorporated into IPPS-A. Furthermore, the system integrator can provide an initial attempt at translating code values based on code value descriptions, if provided. Any proposed updates—including creation, deletion, and modification—relating to domains and domain code sets maintained within IPPS-A are submitted to (and controlled by) the IPPS-A Change Management control process before being implemented.

When data sets (e.g., files, tables, record streams, etc.) are defined as part of interfaces and conversions, any data element (e.g., fields, columns, etc.) that is identified by the trading partner as being restricted to a set of valid code values must be associated with a fully defined domain code set in order to account for the data element's requirement for translation. This means that as part of the data acquisition process for a system's interface and/or conversion data set, that the trading partner must explicitly identify whether there is a list of valid code values for a given data element, and either: where that list is available for review (e.g., TESS ETABLE "ACFTSR"), or the full list of valid code values and their meanings. During this time, Army SMEs will perform the aforementioned analysis as to how the data element's associated code set aligns with existing domains and master values within IPPS-A. Once a data element is associated with a domain code set which is associated with a set of master values under a domain, then translation between that data element's code values and any other related domain code sets can be performed.

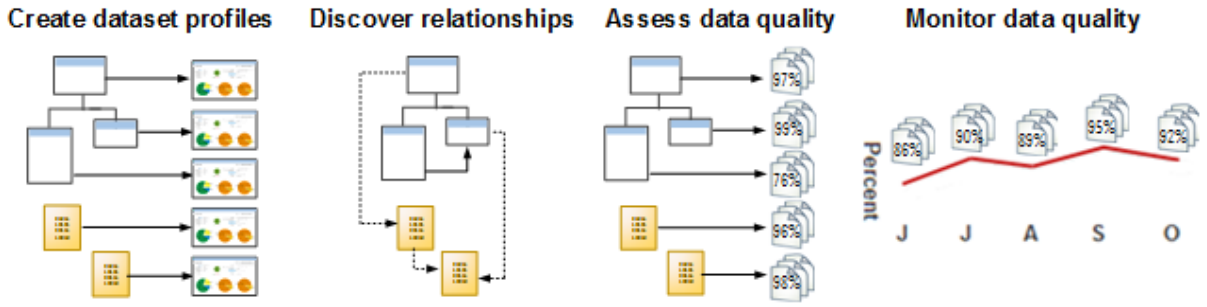
3.2.4 Quality

The primary purpose of the data quality analysis process is to determine the validity and completeness of source data required for conversion and interfacing into IPPS-A to meet the intended business purpose. Building confidence in the IPPS-A enterprise data requires measuring, correcting, and monitoring data quality. These activities will be done using Ab Initio Data Quality Environment.

The Data Quality Environment quality workflow includes the following activities:

- **Create dataset profiles** — Using Data Discovery, files that are landed in the IALZ will have profiles created that have in-depth information on each dataset: summary statistics on each field, whether each field is fully populated, and information about the distribution of values in the dataset.
- **Discover relationships** — Using Data Discovery, as required, datasets will have enhanced profiles computed for groups of datasets, to improve discovery of data relationships between datasets, compare reference data sources, and locate personally identifiable information (PII).
- **Assess data quality** — Using Data Quality Assessment, developers will write tests and validation rules documented in the IDDs that detect data quality issues in order to assess the data quality of each dataset. The assessment consists of scores that measure the percentage of dataset records that are without issues.
- **Monitor data quality** — Using Data Quality Assessment, datasets can be monitored by computing data quality scores and importing these scores to the Metadata Hub. By making successive measurements of a number of datasets and importing them to the Metadata Hub, IPPS-A can monitor enterprise data quality over time. Figure 3-2 depicts the high-level data quality analysis process.

Figure 3-2 High-Level Data Quality Analysis Diagram



The IDD documents the input source data and contains all the business rules used to build the data quality process for field level validations and monitoring. Issues are detected based on the validation tests that get authored from the IDD rules. Data quality metrics are computed when data quality configurations are run each time an interface or conversion is executed. Metrics such as Completeness, Conformity, and Accuracy give the percentage of records that have no issues. Each of the standard metrics has associated issue codes. For example, the Completeness metric has the issue codes AB_ERR_IS_BLANK and AB_ERR_IS_NULL. Data Quality Assessment uses counts of issue codes to compute the percentage of good records — the records that do not contain blank or NULL values. Custom metrics, if identified, can be configured and associated to issue codes. The result sets of running the data quality configurations will be imported to Metadata Hub for visibility to the IPPS-A Triad for ongoing monitoring. In addition to the record level validations that are identified in the IDDs, PeopleSoft Component Interface checks are built into the Ab Initio load graphs. Records that pass through the component-level checks gets classified as "good records" and "bad records". The bad records are written to a reject table in the staging database for analysis and resolution. See Program level Data Quality Tracking SV-10c and Data Quality SOP for resolution process. The good records move on to the next phase, which is the table-level validation.

Similar to the component-level validation, table-level checks are executed to determine whether logic violations exist at the record level. If a logic violation exists, then the error records are rejected and captured in the reject table. The good records move on to being validated at the field level. At the field level, the records are validated for logic violations at field level. The checks for data type, domain values, and completeness are driven by the fields mapped to key or required fields on the target table. The good records move on to the next step in the conversion process, which is the transformation and subsequent migration into IPPS-A.

Reject errors are accessible by the PMO and FMD to analyze and report back to System Owners. Quality improvement is an ongoing process, and the quality of source data received cannot be guaranteed to meet conversion requirements. The data quality process is performed for all PeopleSoft components through standardized Ab Initio component loaders containing the PeopleSoft component specific edits. *Note: sections 4.2.1 Data Quality Analysis and 5. Data Quality Analysis Reports contain additional information.*

3.2.5 Transformation and Migration

The next steps of the data conversion process are transformation and migration. At this stage, records that have successfully passed through the data quality analysis process are transformed (based on the mapping and business rules in the IDD) before finally being loaded into the IPPS-A PeopleSoft HCM database, utilizing the IPPS-A Transformation Services (IATS), see ESB CDRL 07-004 for details. In addition to the business logic required to transform the data, the component loader checks are applied to ensure successful load:

- Component dependency check ensures successful record transformation into the parent component before processing into a child component. One example is a check for existence of a record in the PERSON_DATA component for processing a record into the JOB_DATA component.
- Domain Crosswalk check ensures consistent representation of data values across data sets; this step doubles as an integrity check for domain values.
- Prompt Table check ensures that a transformed field value exists in its associated prompt table. Checking DEPTID from the PS_JOB table against PS_DEPT_TBL is an example.
- Referential Integrity check ensures the referential integrity maintains the transformed PeopleSoft component. An example is checking to ensure that all tables in PERSON_DATA component have the right number of records.
- For complex conversions, Ab Initio will load data into PeopleSoft using Component Interfaces. For simple interfaces that do not require transformation of data or for custom tables, Ab Initio may be used to load directly to the application tables.

3.3 Error Handling

Errors generated during the conversion process are rolled into a Data Quality Error Framework (DQEF) which is used in generating the Data Quality Analysis Reports (DQAR) as described in Table 3-1 and Table 3-2.

Table 3-1 IPPS-A Data Quality Error Framework

Error Type	Description
Accuracy	Value must be in a defined Domain Code set, in a defined lookup file, or in a defined range or list of special values
Coherence	Value is not an outlier
Completeness	Value is not NULL or blank
Conformity	Value is valid for its data type and in the correct format or pattern
Consistency	Value is consistent when compared to a) another value in the same record, b) another value in a different dataset, or c) the previous version of the same dataset
Dataset Integrity	Indicates that the dataset is well formed, including the correctness of header and trailer records
Integrity	Foreign key value matches known primary key and other referential integrity tests
Stability	Distribution of values is consistent with historic distributions

Error Type	Description
Timeliness	A measure of the degree to which the data is available in a timely fashion
Uniqueness	No value occurs more than some configurable value

Table 3-2 depicts the error classes defined in the DQEF.

Table 3-2 IPPS-A Data Quality Error Classes

Error Class	Description
DV	Data Validation- Domain Values Checks, Date Formats, Missing Key Data this is the predominate type of errors
EN	Environment (mostly used as a catchall) issues, such as check sum failures.
PE	Performance – If a file did not arrive on schedule within the defined window, job ran longer than allowed.
TH	Threshold – This tracks drastic change in files size, number values these are generally alerts to possible issues with the data.

Data errors relate to the quality of the source data as received from the Army System Owners.

The Data Quality reports formats and layouts are described in the CDRL 10-002 Data Quality Analysis Reports (DQARs), resolution process is described in the Data Quality SOP.

3.4 Mock Conversions

Mock conversions provide test cycles to build up to full end-to-end data conversion process, including the sequencing, tools and data transformation rules. The mock conversions are intended to reduce risk as the process and programs mature leading up to the go-live event. IPPS-A will perform multiple mock conversions prior to the start of Limited User Testing (LUT). This method allows the SI to validate that the converted data meets the IPPS-A's expectations and accuracy thresholds, and allows the evolution of the conversion process.

The DQEF and DQAR will be utilized to capture and communicate errors and defects encountered in each mock cycle. Defects are logged in Application Lifecycle Management (ALM) and synchronized with the Agile Manager for resolution by the Data Team. Resolution of conversion defects minimizes transformation defects for the inbound interfaces originating from the same source. The defect resolution process with system owners for source data will be defined and conducted by PMO and FMD with support from the SI.

In addition, mock conversions allow IPPS-A to:

- Estimate the data volume of the static data and project the production data volume of the transactional data
- Capture processing times to better estimate environment sizing and time frames required
- Track records accounts and load results as a baseline for each conversion
- Enhance any validation scripts/procedures required to ensure a proper business validation and sign-off for an interface
- Better handle exceptions and errors

- Train the business stakeholders and testers to get familiar with the details of the validation process to be able to execute them prior to cutover
- Prepare the business stakeholders and testers to execute and validate data constructs for interface testing
- Ensure a successful cutover for the interface, infrastructure, application builds, and configuration transports
- Meet the user requirements and adhere to the approved data quality standards

3.5 Disposition of Data

Once the test data files are no longer needed, as directed by the Data Lead, the original files, and any copies, will be deleted from all environments. This is required in order to limit the amount of storage space consumed and the risk of data being exposed in environments that are not authorized for Personally Identifiable Information (PII).

3.6 System Subsumptions

IPPS-A is responsible for the subsumption of 40-plus systems during Increment II. The following artifacts are required prior to release kick-off to enable analysis and development for each system:

- SVs (SV-1, SV-2, SV-3, SV-4, SV-6, and SV-8)
- List of interfaces not listed on the SV-1
- Data dictionary
- User manuals
- Schema of the system
- Source Code of subsumed systems, upon request
- For partial system subsumptions, the physical tables related to the subsumed functionality
- System roadmaps (i.e., Technical, Requirements, and Engineering Change Request [ECR])
- System requirements (i.e., Auditing and Data Archiving)
- File Layouts for all inbounds and/or outbounds of the system (i.e., IRS and/or IRA)
- Domain value codes
- Field mapping interfaces for all inbounds and/or outbounds of the system (documentation)
- Actual data files used in the data exchanges (sample data for each interface)
- Database technology
- Identify the DoD, Dept. of the Army, and other regulations and policies that drive the system requirements
- Relevant Performance requirements (i.e., current SLAs)
- System storage metrics: history, current, and projections

4 INTERFACES STRATEGY

The interfaces strategy provides an overall approach that is used in development of the persistent interfaces in order to integrate IPPS-A with all required external systems. The IPPS-A interface strategy for Release 3 and beyond expands on the capabilities developed in Release 2 and introduces standardized messaging via web services for the exchange of authoritative data with systems that are capable of producing and consuming web services.

4.1 Interfaces Scope

The general scope for interfaces is defined in the IPPS-A's SV-8. The SV-8 is a living document and changes periodically. Based on the SV-8 list of systems, a detailed Work Item/Feature scope is baselined specifying the type of data that is consumed/published from/to the underlying PeopleSoft components. Changes to data Work Items/Features is governed by the Engineering Change Board (ECB) process.

4.2 Inbound Interfaces Approach

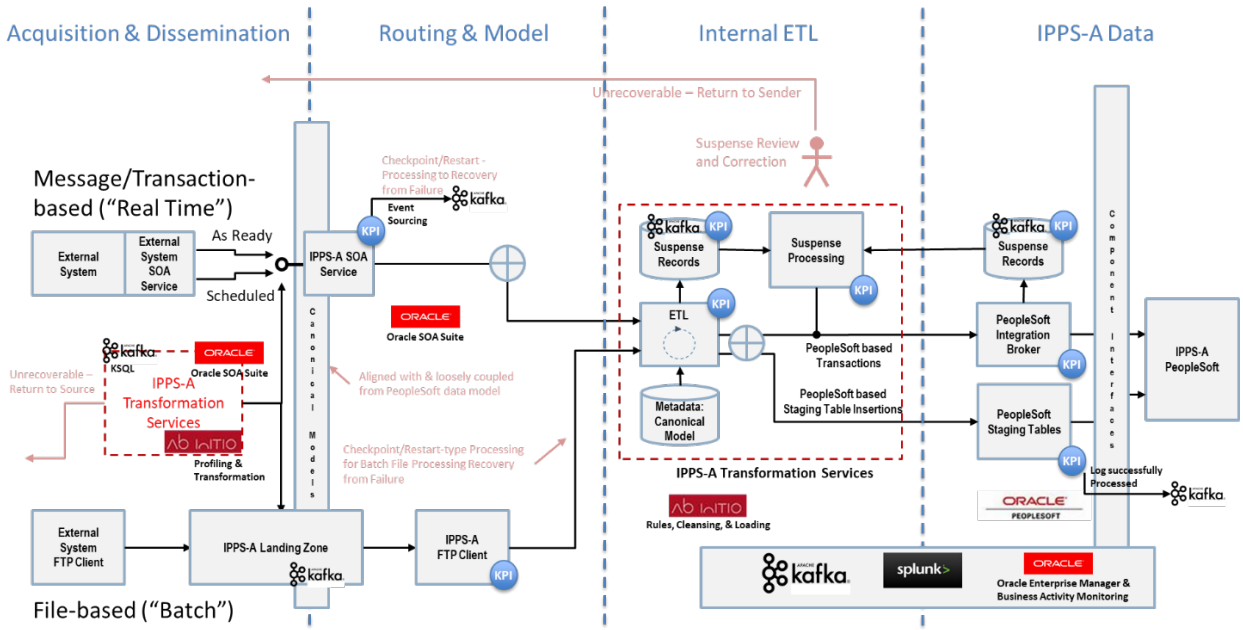
This section describes how data that comes into IPPS-A from external systems is interfaced within IPPS-A. The SOA/ESB integration architecture for Increment II, Release 3 is planned around two primary inbound patterns, file-based and message based, as depicted in the following figure. Additional information on the inbound patters is available in CDRL 07-004, ESB Integration Plan.

As described in the CDRL 07-004, ESB Integration Plan, inbound processing will use a common event sourcing pattern for all inbound flows. Oracle Service Bus (OSB) will provide that capability for inbound service call putting the information to an inbound topic that is grouped in either the member, organization, or organization member grouping. Ab Initio provides inherent tracking capability that surfaces processing information in Control Center, additionally an Apache Kafka adaptor will use a file source adapter to stream this information into the common logs under the same grouping as done for services. In addition to event sourcing, the inbound processing will track Key Performance Indicators (KPIs) identified in IDDs and store them in topics under the same grouping classification.

This tracking of the progress through the steps identified in the conversion approach above will exist for event sourcing and surfacing of KPIs throughout the inbound & outbound flows where Apache Kafka will provide the standard logging framework and the sources will leverage its adapters to surface the information. Business Activity Monitoring (BAM) capabilities from the ESB will be used to alert and inform interface partners of activity resulting from processing inbound and outbound interfaces. This capability includes notification for inbound customers wanting guaranteed delivery that agree to provide a secondary file identifing the SHA-256 checksum. Daily operations of the IPPS-A system, including interface monitoring are documented in the IPPS-A Runbook.

Inbound processing utilizes scheduling tools that allow job sequencing and dependencies to be established. Schedule processing will leverage service orchestration and business rules to expose complex processing and conditional logic branching as required.

Figure 4-1 SOA/ESB Inbound Integration Overview Diagram for Release 3



4.2.1 Data Acquisition

Inbound interfaces utilize the same capabilities and process as described for Conversions to handle data Acquisition, Staging, and Domains.

4.2.2 Data Quality Analysis

The persistent inbound interface data is processed through the same Data Quality Error Framework (DQEF) as the conversion programs a runtime facility. If IPPS-A has both a conversion and an inbound interface from the same source to the same PeopleSoft component, the data quality, mapping and load logic will be captured in separate IDD. Change data capture compares inbound files to current system values to reduce errors and eliminate redundant and duplicative processing. Furthermore, the component loader processing the interface data is loaded via the PeopleSoft Component Interface to ensure that invalid data will not be uploaded into IPPS-A. Data accuracy for conversions and inbound interfaces is calculated as the percentage of authoritative data elements loaded into IPPS-A that reflect the data elements received from the authoritative system. Further information regarding data accuracy requirements can be found in the IPPS-A Increment II, Post-MS B TEMP.

4.2.3 Interfaces Transformation Process and Inserts, Updates, or Deletes

Transformation is the final step in the interface process. Based on the business rules in the IDD, external records or transactions that have successfully passed through the data quality checks and transformations are executed by the IATS and committed into the IPPS-A PeopleSoft Component.

- Inserts – as per the PeopleSoft terminology, true inserts occur when a new level 0 record for a PeopleSoft component is received via the interface. For example, a new soldier record is received from an accession system resulting in the creation of a new PERSON record with a new EMPLID. Inserts to non-level 0 records from a PeopleSoft component perspective are considered updates.
- Updates – as per the PeopleSoft terminology, updates can occur when a level 0 record is updated (for example, DOB is changed for an existing soldier) or when a non-level 0 record is inserted (for example an existing soldier has a new home address). Inactivations and activations of records in PeopleSoft are done via Updates with new status (and often Effective Date, or EFFDT).
- Deletes – as per the PeopleSoft terminology, deletes for level 0 and most of non-level 0 records are not permitted. Although the framework allows for a page to be constructed allowing a non-level 0 record to be deleted, IPPS-A has not identified a scenario in which a delete is required through an inbound interface.

4.2.4 Domains, Domain Code Sets, and Code Translations

The catalog of reference data domains described in section 3.2.3 applies to inbound and outbound interfaces.

4.2.5 Error Handling

Data quality and transformation error handling for inbound interfaces uses the error handling framework described for conversions described in Section 3.3. Error handling for Enterprise Service Bus (ESB) operations is described in CDRL 07-004 ESB Integration Plan. Individual system interface error handling, if required, is defined in the IDDs. Validations conducted by PeopleSoft components for the target fields are identified in Section 8.3 of the interface IDD. System interfaces are identified in the Runbook along with the associated error handling procedures.

4.2.6 Template Design

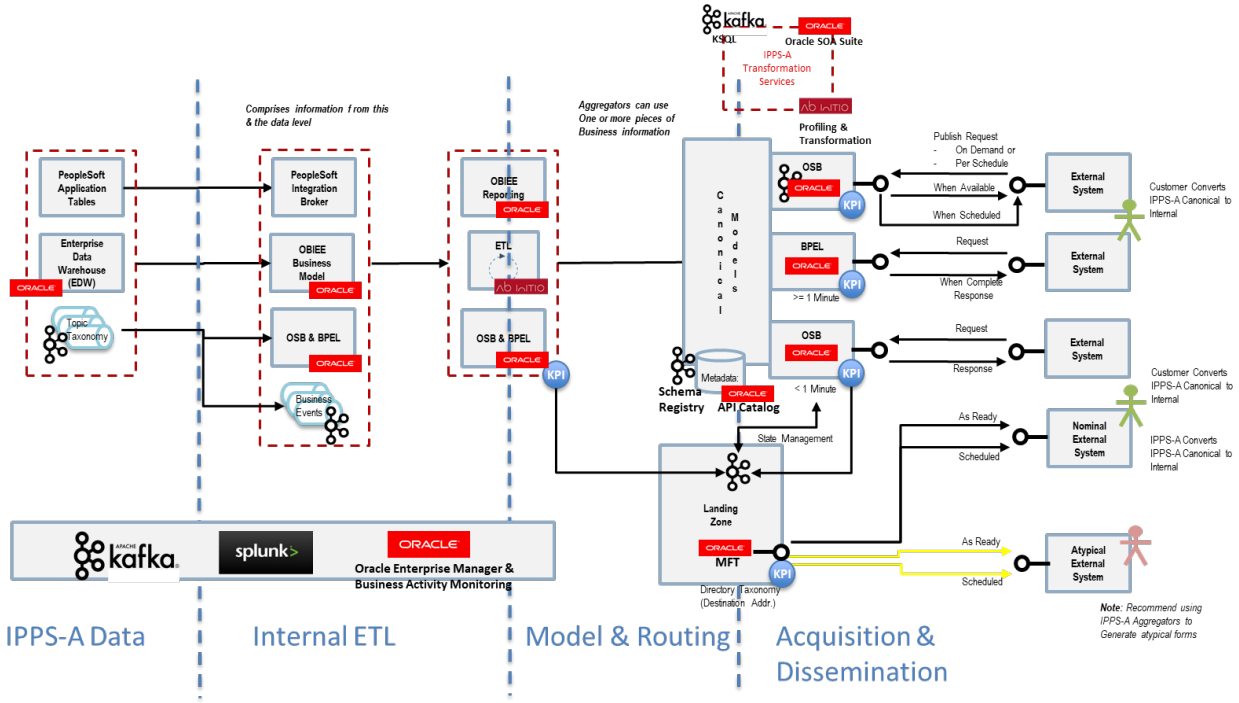
Each interface should include common elements to allow devops to efficiently manage the interface. Elements suggested based on R2 Production Runbook include:

- System Name
  - (ARISS, AOS, etc.)
- Disruption Impact
  - Ex: "Loss of new hires for the processing period."
- Recovery Priority
  - Low, Medium, High, Critical
- Delivery Type
  - Queue, Flat File, Delta, etc.
- Source POCs
  - Technical, Functional, ESB (Name and Email)
- Prerequisites to running
  - Ex. DQ Pre-run setup has completed at least once
  - Ex. ITAPDB Inbound has completed
- File Locations
  - Ex. MFT: /path/to/file
  - Ex. Ab Initio Pending: /path/to/file
- File Interfaces
  - Ex. ATTTRS: UDAU.TRAN.DATA
- System Interface Information
  - Table with columns: Transfer Method, Frequency, Day/Time of Delivery, Load Type, Format
- Queue endpoint information
- Run Steps
  - Include any assumptions such as ARISS queue has been loaded
  - Include any Stonebranch/PS jobs
- Special Processing Considerations
  - Eg. ATRRS files are one delta file per day.
  - Eg. How to handle catch-ups
- Data Archiving
- Execution Frequency
- Supporting Document Links
  - Common document links include: Data Interface Information Form, Interface Design Document, Interconnection Security Agreement, MOA/MOU

4.3 Outbound Interfaces Approach

This section describes how data that flows outbound from IPPS-A to external systems is interfaced. IPPS-A will lead the Army in modernizing how Army HR systems exchange and synchronize soldier data across the enterprise. By utilizing messaging and event driven exchanges of data, changes in HR data will be distributed to consuming systems in an efficient manner. As new consumers of HR data are identified, systems can attach to the IPPS-A ESB and then subscribe to published message Topics in order to receive the required HR data. Figure 4-2 depicts the outbound flow pattern for synchronous and asynchronous services and file transfer. Additional information on the outbound patters is available in CDRL 07-004, ESB Integration Plan.

Figure 4-2 Outbound Flow Pattern Synchronous / Asynchronous Services and File Transfer



4.3.1 Design Principles

IPPS-A's outbound interfaces strategy follows six (6) technical best practices principles for building and maintaining the Army's complex data exchange landscape.

1. **Standardized Layouts.** IPPS-A will publish PeopleSoft component-driven XML messages. XSDs will be based on views (and not physical tables) to minimize the need to republish XSDs due to data model changes caused by application enhancements or vendor updates.

Example: one XSD for all Person Profile Component data.

2. **Standardized Transactions.** Message XMLs will follow the PeopleSoft component functional transactions which are also referred to as topics.

3. **Standardized Domain Values.** IPPS-A has adopted the Army's Authoritative Data Sources (ADS) for its external domain data and consumer systems that use non-ADS domain values will be responsible for maintaining crosswalks. IPPS-A will publish any IPPS-A specific domain values.

Example: IPPS-A will publish Action/Action Reason domain values to allow consumers to correctly interpret IPPS-A Job/Assignment data.

4. **Standardized Delivery Mechanisms.** CDRL 07-004 ESB Integration Plan describes the delivery mechanisms and patterns. IPPS-A's preference is that most consumers will subscribe to IPPS-A data via Publish-Subscribe service, but a file delivery is also available for external systems that do not have ESB capabilities.

Example: System A can receive Personal data message via Publish-Subscribe, System B can receive the same identical message files via Request-Reply, and System C can receive by SFTP via Oracle MFT transfer.

5. **Standardized Agreements.** Standardized data and message exchanges will enable the PMO to sign and maintain MOAs and ISAs with standardized message exchange characteristics.

Example: two consumers subscribing to soldier Personal data can have identical data structure and use descriptions.

On a limited basis, IPPS-A will follow a formal review through the Engineering Change Board process to address requirements for accommodating point-to-point outbound interfaces based on specific legacy layouts and/or domain values. The SI recommends that such exceptions be kept to a minimum in order to limit any schedule and risk impact.

4.3.2 Message Delivery Mechanisms

CDRL 07-004 ESB Integration Plan describes the IPPS-A message delivery patterns.

4.3.3 Documentation

CDRLs for Outbound Design Document, and individual system IDDs contain additional information on topic based exchanges and point to point designs.

4.4 Interface Trial Runs

Interface trial runs allow the Data Team to test the interface processes. This iterative process provides an opportunity to verify the processed data and make the necessary adjustments to the processes prior to the interface going live. In addition, trial runs allow a more accurate assessment of the actual time necessary for each of the interfaces to run.

The expectation of the Data Team is to perform multiple trial runs to ensure that each of the interface programs is working as specified in the IDD, and meets the performance thresholds and all requirements. Each testing cycle requires the Data Team and the Functional Team to perform validation of the transformed data.

5 DATA QUALITY ANALYSIS REPORTS

The automated Data Quality Analysis Reports (DQARs) are provided as part of CDRL 10-002. The SI recommends that End-User Verification of Data is performed on all converted data as early as possible during the mock conversions and trial interface runs. This will allow the end users to ensure that the data in IPPS-A meets the Army's functional needs.

As described in section 3.3, Errors generated during the conversion process are rolled into a Data Quality Error Framework (DQEF) which is used in generating the Data Quality Analysis Reports (DQAR).

6 DATA CONVERSION AND INTERFACES TOOLS

The tools listed below can be leveraged for IPPS-A data conversions and interfaces:

- Ab Initio Toolset (Graphical Development Environment [GDE], Express>It, Query>It, Test Data Management, and Enterprise Meta Environment [EME])
- PeopleSoft Application Engine
- PeopleSoft Integration Broker
- PeopleSoft Component Interface
- SOA/ESB

6.1 Ab Initio

Ab Initio's technology is a single, scalable, unified platform built from the ground up for highly parallel and efficient data processing (including data tables, files, message queues, Web services, and metadata). Ab Initio provides easy to use, robust graphical and rules-based interfaces to handle complex data cleansing issues and migration logic.

The data integration process using Ab Initio supports error handling, data cleansing, and subsequent data quality testing in multiple measurable phases, providing visibility across the enterprise to support effective decision making.

Ab Initio's Metadata Hub includes an inventory of current source systems metadata, target metadata, systems information, data structure, relationships, domain values, and legacy data anomalies that require cleansing. This tool simplifies and accelerates both data synchronization and consolidation; it also streamlines the building of validation rules and mappings that drive IPPS-A data cleansing and conversion/migration. Data lineage will be available in Metadata Hub for all analysis-ready code and datasets, including Ab Initio executables, Oracle database tables, and vendor-exposed transformations and data structures within PeopleSoft.

6.2 PeopleSoft Integration Broker

PeopleSoft Integration Broker is a middleware technology that performs asynchronous and synchronous messaging among internal systems and third-party systems. It exposes PeopleSoft business logic as Web services to PeopleSoft and third party systems. It also consumes and invokes third-party Web services from PeopleSoft systems.

PeopleSoft Integration Broker enables integrations among internal systems and external integration partners including an ESB or ETL tools, while managing data structure, data format, and transport disparities. Integration Broker consists of two subsystems: the Integration Gateway and the Integration Engine. The Integration Gateway resides on a PeopleSoft web server, and the Integration Engine resides on an application server as part of the PeopleSoft application.

6.3 PeopleSoft Application Engine

The PeopleSoft Application Engine is a tool designed to develop batch-processing types of programs. The Application Engine program is a set of Structured Query Language (SQL)

statements, PeopleCode, and program control actions that enable looping and conditional logic. The program creates through the PeopleSoft Application Designer and performs a business process. The PeopleSoft Application Engine has the option for straight, row-by-row processing, but the most efficient Application Engine programs write to perform set-based processing.

The Application Engine program utilizes the same edits and PeopleCode logic built into the online PeopleSoft applications, thereby ensuring processing consistency and data integrity.

6.4 Component Interfaces

Component Interfaces apply the component edits during the upload of external data into PeopleSoft. The best use for Component Interfaces are when there is a relatively small dataset and complex Component edits.

6.5 SQL/DMS Scripts

SQL is a standard data manipulation language delivered as part of the Oracle database. PeopleSoft Data Mover Scripts (DMS) are a high-speed data manipulation tool that facilitates data transfer within PeopleSoft databases.

6.6 SOA/ESB

Service Oriented Architecture (SOA) helps transform complex application integrations into Agile and reusable service-based connectivity to speed system integrations, and respond faster to business requirements.

SOA separates business functions into distinct units, or services. A SOA-based solution reuses services to automate a business process. A standard interface and message structure defines services. The most widely used mechanisms are Web-based standards including the Web Service Description Language (WSDL) file for service interface definition and XML Schema Documents (XSD) for message structure definition. These XML standards are easily exchanged using standard protocols. Because standards for Web services use a standard document structure, they enable existing systems to interoperate regardless of the choice of operating system and programming language used for service implementation.

IPPS-A has selected the Oracle SOA platform as the IPPS-A ESB solution. The platform includes the core products of Oracle SOA Suite, Oracle Service Bus (OSB), and Oracle Managed File Transfer (MFT).

In addition to the Oracle SOA products, Apache Kafka is incorporated into the overall SOA stack to provide the low latency platform for handling the near real time outbound data feeds.

7 CONFIGURATION MANAGEMENT

This section addresses Ab Initio procedures for interfaces and conversions, see the Configuration Management Plan CDRL 13-009, for full details of the system wide plan. The configuration management for conversion and interface code in Ab Initio uses versioning and tags. Ab Initio employs a check-in/check-out process to maintain source control. All objects in the Ab Initio technical repository have controlversion numbers associated with them. A version number is an integer that is incremented every time a user checks an object into the repository. This versioning allows developers to revert to earlier versions of code and manipulate specific versions of the code if required. In addition, it also enables developers to compare different versions of the same object and highlight differences between them. A tag in Ab Initio is a string of text that enables objects to be grouped and referenced as a unit. The most common use of tags is to identify and group objects in a given release.

Conversion and interfaces leverage this capability of Ab Initio for configuration management. All graphs include a version number and additional attributes such as design document (IDD with version number) and other related information. Tags use these objects to group at the increment/release level.

Appendix A: Acronyms and Abbreviations

Acronym	Definition
ADS	Authoritative Data Source
ALM	Application Lifecycle Management
API	Application Programming Interface
BBS	Batch Broker Service
BAM	Business Activity Monitoring
BI	Business Intelligence
CDRL	Contract Data Requirements List
CHRIS	Common Human Resources Information Standard
COTS	Commercial Off-the-Shelf
DCIP	Data Conversion and Interfaces Plan
DMS	Data Mover Scripts
DOB	Date of Birth
DoD	Department of Defense
DQAR	Data Quality Analysis Report
DQEF	Data Quality Error Framework
ECB	Engineering Change Board
ECP	Engineering Change Proposal
EME	Enterprise Meta Environment
ESB	Enterprise Service Bus
ETL	Extract Transform Load
GDE	Graphical Development Environment
HCM	Human Capital Management
HR	Human Resources
IDD	Interface Design Document
IPPS-A	Integrated Personnel and Pay System – Army
IRS	Interface Requirements Specifications
ISA	Interconnection System Agreement
JMS	Java Message Service
MFT	Managed File Transfer
MOU	Memorandum of Understanding
PMP	Project Management Plan
SFTP	Secure File Transfer Protocol

Acronym	Definition
SI	System Integrator
SOA	Service Oriented Architecture
SQL	Structured Query Language
SSN	Social Security Number
TESS	Table Edit Support System
UIC	Unit Identification Code
XML	Extensible Markup Language
XSD	XML Schema Definition

Appendix B: CACI Consolidated Reference Documents

This appendix lists other referenced documents that are provided to complement required Contract Data Requirements List (CDRL) items.

Document #	PWS Reference	Document Title
01-004	4.1.4	Project Management Plan (PMP)
05-011	4.5.7	Outbound DD
07-001	4.7.1	Interface Design Document (IDD)
07-004	4.7.4	ESB Integration Plan
13-005	4.13.1	Interface Requirements Specification (IRS)

Appendix C: Handling of PII Data

I. Background

The SI has been requested to scramble PII to protect sensitive soldier data. The SI will scramble the following elements:

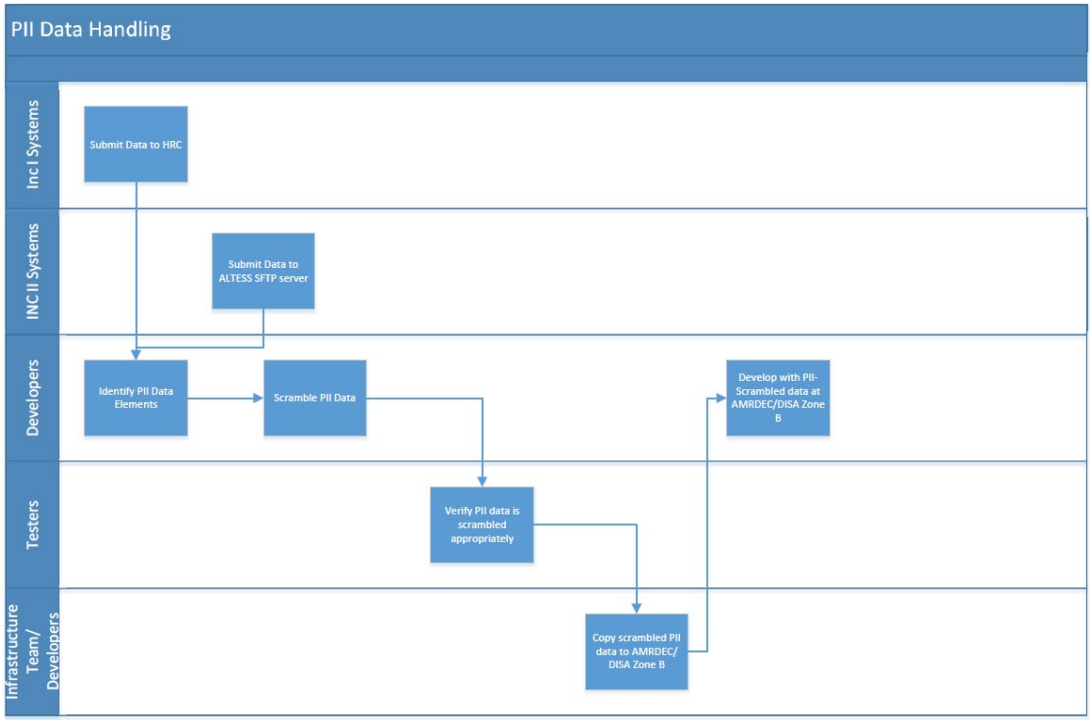
- Name
- Address (of a person, not of Government facilities)

NOTE: the state and zip code of a person's address will not be scrambled.

- Social Security Number (SSN)/Driver's License Number
- Phone
- Email address
- Photo

The Government has requested the SI restrict PII data access to personnel whose job responsibilities dictate the need for and for the time required. PII data will be scrambled in DISA Zone A and the SI will restrict access to only personnel conducting data quality checks and masking PII data, with no partial access granted. The SI should not need any additional configuration tasks beyond revoking the accounts mentioned above once tasking is complete. Figure C-1shows how the SI will handle PII data.

Figure C-1 PII Data Handling



The masking approach will be as follows:

For any PII data element that is identified as a join key between datasets (most commonly SSN), the masking algorithm will permute the characters in the data in a consistent way, so that when the two identical inputs are masked, they will produce identical outputs that are different from the input. For instance, given an input SSN of 123456789, the masked output might be 548672315. For all instances of SSN 123456789, across all systems and records, the masked value would always be 548672315, so that the records can be joined across datasets. The permutation function takes two inputs: the data to be masked, and a cipher key. The cipher key is used as a seed for the permutation function, and the same cipher key must be used for each SSN that is masked in order for the output to be consistent. The cipher key has a default value that will be changed to a random text string to prevent attempts to build a crosswalk of input values to output values for all valid SSN combinations based on the default cipher key.

For any PII data element that is not identified as a join key (such as address, name, etc.) the data will be masked by assigning fictitious data in the form of either random data, a fixed fictitious value, or a fictitious value selected from a list of pre-defined fictitious values. For instance, if a record had a name field with the value of John Doe unmasked, the masked value could look like:

- Random: fmblz
- Fixed fictitious value: Robert Smith (for all names, all records)
- Fictitious value from a list: records might randomly get one of George Washington, Abraham Lincoln, etc. based on a list of names.

Testers will verify that the input name never matches the output name (in cases where a person's actual name was the same as one of the fictitious names, it would be re-run with a different random seed, or manually corrected prior to approval).

II. DISA ZONE A

ESB is implemented which transports data from source systems and routes files to an encrypted file system for further processing.

In case of outbound interfaces, DISA Zone A PII data should only interface with external systems that have accreditation to handle PII. The SI baseline development effort does not provision for masking of outbound data – functionality can be implemented on a pre-system basis if requested by the Government.

III. DISA ZONE B

The SI will copy the scrambled PII data over from DISA Zone A to Zone B. The SI Developers will use DISA Zone B to continue build IPPS-A with scrambled PII data.

IV. PII Environment - DISA Zone A

Once source datasets have been decrypted onto an encrypted file system, the data will be processed for quality issues. Each dataset containing PII elements will go through a scrambling process that will transform the PII elements in a non-reversible way. The SI requires that the

source systems identify the fields in the source dataset that contain PII information for scrambling.

The SI will maintain a crosswalk between the PII-scrambled data and the PII data so that any issues discovered while processing the PII-scrambled data can be traced back to the original records and reported to the system owners. This crosswalk will exist only in the PII environments and will not be placed in other environments. PMO and FMD personnel responsible for review and adjudication of data load errors will have access to the crosswalk following the applicable security policies.

The SI will produce quality reports for PII data separate from non-PII data quality reports. Access to UNIX, and the Express>It/Metadata Hub (managed by SI Ab Initio administrators) will be restricted to only users who have been assigned data quality or PII masking tasks (as well as Infrastructure team, etc.).

V. ACCEPTANCE

The SI plans to execute based on the above outlined PII-handling procedures as per the build schedule. The SI will incorporate additional requirements or blockers as the Government presents them.

Appendix D: R3 Conversion Strategy

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